A Sequential Approach for Accurate Parameters Identification of Heavy-Duty Hydraulic Manipulators Ensuring Physical Feasibility

Weidi Huang, Zhiwei Chen, Fu Zhang, Min Cheng, Ruqi Ding, Junhui Zhang, and Bing Xu

***Abstract*—Accurate identification of dynamic parameters is essential for precise motion control and autonomous operation of heavy-duty hydraulic manipulators. However, due to their low-speed motion property, conventional approaches fail to simultaneously excite all parameters. To overcome this issue, a sequential parameter identification approach for heavy-duty hydraulic manipulators is proposed. All parameters are categorized based on their dynamic characteristic, and then distinct excitation trajectories have been designed to separately stimulate and identify each parameter. Dynamic parameters are fully excited, which is reflected in a reduced condition number of the observation matrix. Furthermore, an approach that ensures the physical feasibility of the identified parameters is constructed, which makes them more suitable for application in nonlinear control. The performance of the proposed method is evaluated with various identification methods, including traditional least squares, weighted least squares, and the method only considering physical feasibility. The results indicate a substantial decrease in torque prediction error compared to these methods. Specifically, the prediction accuracy of the joint torque using the proposed method has been improved by approximately 5.63% to 27.06%.**

***Index Terms*—Hydraulic manipulators, parameter identification, physical feasibility.**

I. Introduction

Hydraulic manipulators known for their superior power-to-weight ratio and durability, are critically important across a variety of industries, including forestry, manufacturing, construction, and material handling [1], [2], [3], [4]. With the rising demand of the autonomous operation, it requires manipulators to exhibit better performance in terms of the overall efficiency, reliability and safety [5]. A key aspect of improving these performances is the accurate identification of dynamic parameters, such as mass, inertia, friction coefficients and interaction parameters [6], [7]. This process is conducive to establishing an accurate dynamic model, which enables more accurate prediction of the system behavior during the complicated autonomous operational task.

For conventional rigid manipulators, dynamic parameters identification typically involves a rigorous analysis of joint movements and torque signals over a predetermined period [8]. This process is carried out while the manipulator follows a trajectory designed to stimulate various aspects of its dynamic model [9]. Discrepancies in motion and torque are then used to refine the estimated parameters [10], [11].

The most common technique for offline dynamic parameters identification is the inverse dynamic approach, which is usually achieved using least-squares estimation [12] or its derivative approaches [13]. A critical step in these approaches is the optimization of excitation trajectories [14]. To sufficiently excite dynamic properties of manipulators, the condition number of the observation matrix is generally selected as the optimization objective [6], [15]. Despite the effectiveness of these approaches, they are prone to noise and heavily dependent on the specific condition of the robot's trajectory [16]. Particularly, a sufficiently large change in velocity or acceleration is required to fully excite each inertial parameter.

The parameters identification of heavy-duty hydraulic manipulator poses two unique challenges [17]. Firstly, joint torques computed from pressure signals, are particularly vulnerable to noise interference [18]. Secondly, joint velocity is limited by both the pump flow rate and its heavy-load capacity [1]. Therefore, they cannot execute the high-velocity motions that are significant for adequately exciting the dynamic parameters [19]. Insufficient excitation makes the observation matrix become ill-conditioned (the condition number tends to increase), which amplifies the impact of noise on the results. This in turn can cause the identification results to deviate from the true values, leading to a reduction in the accuracy of the predicted torque [6], [20], [21], [22], [23].

When the manipulator is in a stationary state, the joint torque is primarily affected by gravity and friction forces. To reform conventional approaches that simultaneously identify inertia, gravity, and friction terms [17], [18], [24], it has revealed that the independent identification of the gravity term in a stationary

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state can yield superior results [25], [26], because the dominant influence of the gravity term on the load torque can be fully excited. Moreover, by utilizing force sensors, the effects of friction can be effectively eliminated [27], [28], allowing for the separate identification of gravity and inertia terms, and also enhanced force control and load estimation can be achieved [29], [30]. However, a substantial limitation of this approach is its dependence on force sensors, leading to increased cost, complex installation processes, and higher maintenance requirements.

Summarily, the key challenge in parameter identification of hydraulic manipulators is simultaneously exciting both the dynamic and friction parameters sufficiently. During low-speed motion, the degree of excitation of different dynamic parameters is not consistent, which leads to the ill-conditioned observation matrix and affects the identification results. For hydraulic manipulators, due to its properties of low-speed and heavy-duty, torques used for overcoming gravity usually play a dominant role in joint driving torques compared to the inertial and friction torques. The gravity term is unrelated to velocity variations, while the inertia and friction terms are sensitive. Therefore, a sequential method is proposed for identifying inertia, gravity, and friction parameters separately to fully excite dynamic parameters. The main contributions are summarized as follows:

1) A sequential parameter identification approach is proposed by dividing dynamic parameters into two groups based on their excitation characteristics. One group (the combination of gravity term, Coriolis force term, and centrifugal force term) is excited using a constant velocity trajectory to reduce the effects of friction. The other group (the inertia term) is excited using a combination of finite Fourier series and polynomials. In this way, the decrease in the condition number of the observation matrix for two groups mitigates the difficulty of simultaneously exciting all parameters during slow movements of heavy-duty hydraulic manipulators.
2) The sequential identification method may result in dynamic parameters being unconstrained by physical feasibility. To overcome this issue, the physical feasibility set of dynamic parameters will be updated after each identification iteration, which is different from identifying all parameters simultaneously. Then, a new physical feasibility set of all parameters is obtained, as given in Section III-C.

II. System Modeling

A. Manipulator Modeling

For a n-degree-of-freedom (DOF) hydraulic manipulator as shown in Fig. 1, joint torques are calculated using pressure sensors. Incorporating the most adopted friction force model and based on the principle of virtual work, the joint torque of the hydraulic manipulator can be expressed as

$$\boldsymbol{T}_{au}(i) = \boldsymbol{J}_h(i,i)\boldsymbol{P}_L(i) - \boldsymbol{f}(i), \tag{1}$$

$$\boldsymbol{f}(i) = fv_i\dot{\boldsymbol{q}}(i) + fc_i sign(\dot{\boldsymbol{q}}(i)) + fo_i, \tag{2}$$

where $\boldsymbol{T}_{au}$ is the joint torque, $\boldsymbol{J}_h$ is the conversion coefficient defined in [23], $\boldsymbol{P}_L$ is the acting force if the ith actuator is linear cylinder, $\boldsymbol{P}_L$ is the acting torque if the ith actuator is swing cylinder, $\boldsymbol{f}$ is the joint friction torque, fv_i is the damping friction coefficient, fc_i is the coulomb friction coefficient, fo_i is the disturbance term of the friction.

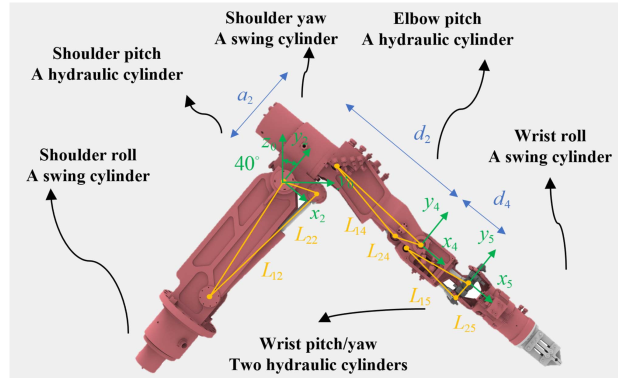

Fig. 1. The studied hydraulic manipulator, n=7.

The dynamic equation of the manipulator is written as

$$\boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{C}\dot{\boldsymbol{q}} + \boldsymbol{G} = \boldsymbol{T}_{au}, \tag{3}$$

where $\boldsymbol{M}$ is the mass matrix, $\boldsymbol{C}$ is the Coriolis and centripetal coupling matrix, $\boldsymbol{G}$ is the gravity matrix.

Note that although the fluid compressibility and valve nonlinearities may cause a phase lag between the measured pressure and the actual actuator response, the heavy-duty hydraulic manipulator investigated here moves at low-speed conditions with relatively small pressure fluctuations, making the torque error introduced by these factors negligible. Therefore, the complex pressure dynamics of the hydraulic system are not explicitly modeled in this study.

To achieve the parameter identification based on the inverse dynamic and reduce the computational complexity, the dynamic model can be reformulated into a linear regression form. And certain assumptions must be made to ensure the effectiveness and solvability of the model [25]. Firstly, parameters of the manipulator are considered to be time-invariant, ensuring the reliability of estimated values. Secondly, signals such as joint angle, velocity, acceleration, and torque, can be precisely measured, or the measurement noise obeys the random distribution with zero-mean (e.g., Gaussian distribution). It allows least-squares-based methods to provide an unbiased estimation of parameters. Then, (3) can be appropriately transformed as

$$\boldsymbol{Y}\boldsymbol{\pi} = \boldsymbol{T}_{au}, \tag{4}$$

where $\boldsymbol{Y}$ is the regression matrix, $\boldsymbol{\pi}$ is the standard inertial parameters.

B. Physical Feasibility Set of Standard Inertial Parameters

The standard inertia parameters belong to a subset of the real number set $\mathbb{R}^{10n}$, as they need to satisfy constraints of physical feasibility. The physical feasibility constraints can be expressed as the linear matrix inequality

$$\boldsymbol{D}_i(\boldsymbol{\pi}) = \begin{bmatrix} \frac{1}{2}Tr(\boldsymbol{I}_i)\mathbf{1}_3 - \boldsymbol{I}_i & [h_{x,i}, h_{y,i}, h_{z,i}]^T \\ [h_{x,i}, h_{y,i}, h_{z,i}] & m_i \end{bmatrix} \succ 0, \tag{5}$$

where $\boldsymbol{I}_i$ refers to the inertia tensor relative to the joint coordinate system, $[h_{x,i}, h_{y,i}, h_{z,i}]$ refers to the first-order moment relative to the joint coordinate system, m_i represents the mass of the link, $Tr(\cdot)$ is a function that computes the trace of a matrix, $\mathbf{1}_3$ is a three-dimensional identity matrix.

The linear matrix inequalities of n links can be extended into a single linear matrix inequality $\boldsymbol{D}(\boldsymbol{\pi})$ as

$$\boldsymbol{D}(\boldsymbol{\pi}) = diag\left[\boldsymbol{D}_1(\boldsymbol{\pi}_1), \boldsymbol{D}_2(\boldsymbol{\pi}_2), \ldots, \boldsymbol{D}_n(\boldsymbol{\pi}_n)\right] \succ 0. \quad (6)$$

The set $\mathcal{D}$ of standard inertia parameters that satisfy the physical feasibility constraints can be defined as

$$\mathcal{D} \equiv \{\boldsymbol{\pi} \in \mathbb{R}^{10n} : \boldsymbol{D}(\boldsymbol{\pi}) \succ 0\}. \quad (7)$$

C. Physical Feasibility Set of Base Inertial Parameters

In set $\mathcal{D}$, only base inertial parameters have an impact on the torque, so only these parameters can be identified. Therefore, it is necessary to construct a set of physically feasible base inertial parameters. The regression matrix $\boldsymbol{Y}$ can be divided into linearly independent columns $\boldsymbol{Y}_{ind}$ and linearly dependent columns $\boldsymbol{Y}_d$ by the permutation matrix $\boldsymbol{P} = [\boldsymbol{P}_{ind} \quad \boldsymbol{P}_d]$ as

$$[\boldsymbol{Y}_{ind} \quad \boldsymbol{Y}_d] = \boldsymbol{Y}\boldsymbol{P}, \quad (8)$$

where $\boldsymbol{P}_{ind}$ and $\boldsymbol{P}_d$ represent the projection matrix of the linearly independent and dependent columns, respectively. Correspondingly, the independent inertial parameter $\boldsymbol{\pi}_{ind}$ and the dependent inertial parameter $\boldsymbol{\pi}_d$ can be written as

$$\left[\boldsymbol{\pi}_{ind}^T \quad \boldsymbol{\pi}_d^T\right]^T = \boldsymbol{P}^T\boldsymbol{\pi}, \quad (9)$$

(4) can be written as

$$[\boldsymbol{Y}_{ind} \quad \boldsymbol{Y}_d]\left[\boldsymbol{\pi}_{ind}^T \quad \boldsymbol{\pi}_d^T\right]^T = \boldsymbol{T}_{au}. \quad (10)$$

The linearly dependent columns can be expressed as a linear combination of the linearly independent columns as

$$\boldsymbol{Y}_d = \boldsymbol{Y}_{ind}\boldsymbol{K}_d, \quad (11)$$

where $\boldsymbol{K}_d$ is the linear transformation matrix.

Linearly independent standard inertial parameters and linearly dependent standard inertial parameters can be reorganized into base inertial parameters as

$$\boldsymbol{\pi}_b = \boldsymbol{\pi}_{ind} + \boldsymbol{K}_d\boldsymbol{\pi}_d = \boldsymbol{K}\boldsymbol{\pi}, \quad (12)$$

where $\boldsymbol{K} = \boldsymbol{P}_{ind}^T + \boldsymbol{K}_d\boldsymbol{P}_d^T$. The solution of standard inertial parameters from base inertial parameters is indeterminate, as linearly dependent inertial parameters cannot be determined. Linearly dependent inertial parameters are defined as variables, whose transformations can be written as

$$\boldsymbol{\pi} = \boldsymbol{P}\begin{bmatrix}\boldsymbol{\pi}_{ind}\\ \boldsymbol{\pi}_d\end{bmatrix} = \boldsymbol{P}\begin{bmatrix}\mathbf{1} & -\boldsymbol{K}_d\\ \mathbf{0} & \mathbf{1}\end{bmatrix}\begin{bmatrix}\boldsymbol{\pi}_b\\ \boldsymbol{\pi}_d\end{bmatrix}. \quad (13)$$

The set $\mathcal{B}$ of base inertial parameters that satisfy physical feasibility can be written as

$$\mathcal{B} \equiv \{\boldsymbol{\pi}_b \in \mathbb{R}^{n_b} : \exists\boldsymbol{\pi} \in \mathcal{D} \mid \boldsymbol{K}\boldsymbol{\pi} = \boldsymbol{\pi}_b\}, \quad (14)$$

where n_b is the number of base inertial parameters.

The introduction of the inverse transformation relationship in the set can be written as

$$\boldsymbol{D}_b(\boldsymbol{\pi}_b, \boldsymbol{\pi}_d) \equiv \boldsymbol{D}(\boldsymbol{\pi}). \quad (15)$$

The semi-algebraic set $\mathcal{B}$ can be written as

$$\mathcal{B} \equiv \{\boldsymbol{\pi}_b \in \mathbb{R}^{n_b} : \exists\boldsymbol{\pi}_d \in \mathbb{R}^{n_d} \mid \boldsymbol{D}_b(\boldsymbol{\pi}_b, \boldsymbol{\pi}_d) \succ 0\}, \quad (16)$$

where n_d is the number of dependent parameters.

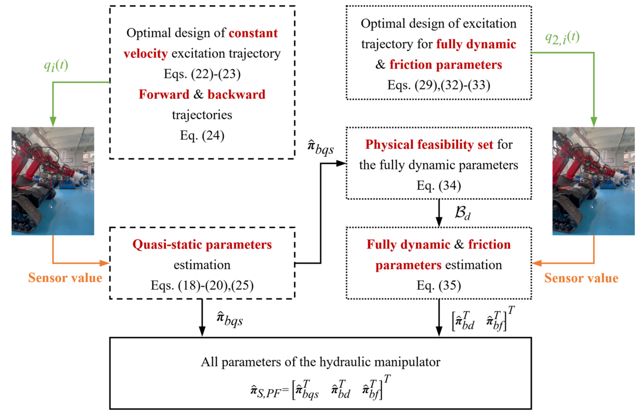

Fig. 2. Flowchart of the sequential parameter identification method.

III. Sequential Parameter Identification

A. Sequential Parameter Identification

Several concepts are firstly defined to better describe the sequential parameter identification method.

Definition 1: Quasi-static parameters represent base inertial parameters that affect the joint torque during the constant velocity motion of the manipulator joints.

Definition 2: Fully dynamic parameters represent base inertial parameters that affect the joint torque only during the acceleration and deceleration motion of the manipulator joints.

Definition 3: Forward trajectory are the trajectory used to excite the quasi-static parameters.

Definition 4: Backward trajectory are the trajectory with the opposite timing compared to forward trajectory and are used to eliminate the effects of friction forces.

The sequential identification method that ensures physical feasibility is proposed as shown in Fig. 2. Initially, quasi-static parameters are identified, and the corresponding constant velocity excitation trajectory is optimized based on the observation matrix of these parameters. Subsequently, fully dynamic parameters and friction parameters are identified using the semidefinite programming approach, and the corresponding excitation trajectory is optimized based on the regression matrix of these parameters. Furthermore, the physical feasibility set of fully dynamic parameters is presented based on the estimated quasi-static parameters.

B. Identification of Quasi-Static Parameters

The excitation trajectory for quasi-static parameters uses a constant velocity trajectory, which can avoid complex friction effects and does not excite fully dynamic parameters. Under a constant velocity excitation trajectory, the dynamic equation of the manipulator can be simplified as

$$\boldsymbol{Y}_{bqs}\boldsymbol{\pi}_{bqs} = \boldsymbol{J}_h\boldsymbol{P}_L - \boldsymbol{f}, \quad (17)$$

where $\boldsymbol{Y}_{bqs}$ is the regression matrix of the quasi-static parameters, $\boldsymbol{\pi}_{bqs}$ is the quasi-static parameters.

The friction force of the hydraulic cylinder is related to the direction and velocity of movement. Thus, the torque of the backward trajectory is collected to compensate for the friction effect, and the dynamic equation can be written as

$$\boldsymbol{Y}_{bqs}\boldsymbol{\pi}_{bqs} = \frac{\boldsymbol{J}_h\boldsymbol{P}_{Lf} + \boldsymbol{J}_h\boldsymbol{P}_{Lb}}{2}, \tag{18}$$

where $\boldsymbol{P}_{Lf}$ and $\boldsymbol{P}_{Lb}$ represent the output force of the hydraulic cylinder for the forwardand backward trajectory, respectively.

The dimension of the observation matrix depends on the number of parameters being identified and the number of data points collected. The observation matrix is composed of a regression matrix of s collected data. The observation matrix for s rounds of data collection can be written as

$$\boldsymbol{\Phi}_{bqs} = \begin{bmatrix}\boldsymbol{Y}_{bqs,1}^T & \boldsymbol{Y}_{bqs,2}^T & \cdots & \boldsymbol{Y}_{bqs,s}^T\end{bmatrix}^T. \tag{19}$$

The sampled torque signal $\boldsymbol{\omega}_{bqs}$ can be written as

$$\boldsymbol{\omega}_{bqs} = \begin{bmatrix}\frac{\boldsymbol{J}_{h1,1}\boldsymbol{P}_{Lf1,1}+\boldsymbol{J}_{h1,1}\boldsymbol{P}_{Lb1,1}}{2}\\ \frac{\boldsymbol{J}_{h1,2}\boldsymbol{P}_{Lf1,2}+\boldsymbol{J}_{h1,2}\boldsymbol{P}_{Lb1,2}}{2}\\ \vdots\\ \frac{\boldsymbol{J}_{h1,s}\boldsymbol{P}_{Lf1,s}+\boldsymbol{J}_{h1,s}\boldsymbol{P}_{Lb1,s}}{2}\end{bmatrix}. \tag{20}$$

The regression equation can be written as

$$\boldsymbol{\omega}_{bqs} = \boldsymbol{\Phi}_{bqs}\boldsymbol{\pi}_{bqs}. \tag{21}$$

To fully excite quasi-static parameters without exciting fully dynamic parameters, the excitation trajectory is designed as a constant velocity trajectory. The analytical expression of the constant velocity trajectory can be written as

$$q_{1,i}(t) = q_{1,i,0} + \frac{q_{1,i,f} - q_{1,i,0}}{T}t,\ i = 1,\ldots,n, \tag{22}$$

where $q_{0,1,i}$ and $q_{f,1,i}$ respectively represent the initial and final position of the excitation trajectory, and T is the period of the trajectory. The variables $q_{0,1,i}$ and $q_{f,1,i}$ are optimization variables, while T is a given parameter. The optimization objective is to minimize the condition number of the observation matrix for quasi-static parameters

$$\text{minimize } \|\text{cond}(\boldsymbol{\Phi}_{bqs}(\boldsymbol{q},\dot{\boldsymbol{q}}))\|. \tag{23}$$

The backward excitation trajectory is generated from the optimized excitation trajectory, and the complete excitation trajectory can be expressed as

$$q_i(t) = \begin{cases} q_{1,i}(t), 0 \le t < T\\ q_{1,i}(2T-t), T \le t < 2T\end{cases} i = 1,\ldots,n. \tag{24}$$

The estimation of quasi-static parameters using the least squares method can be written as

$$\hat{\boldsymbol{\pi}}_{bqs} = (\boldsymbol{\Phi}_{bqs}^T\boldsymbol{\Phi}_{bqs})^{-1}\boldsymbol{\Phi}_{bqs}^T\boldsymbol{\omega}_{bqs} \tag{25}$$

where $\hat{(\cdot)}$ represents the estimated value.

C. Identification of Fully Dynamic and Friction Parameters

Assuming that quasi-static parameters are known, the dynamic equation of the manipulator can be represented as

$$\boldsymbol{Y}_{bd}\boldsymbol{\pi}_{bd} + \boldsymbol{Y}_{bf}\boldsymbol{\pi}_{bf} = \boldsymbol{J}_h\boldsymbol{P}_L - \boldsymbol{Y}_{bqs}\boldsymbol{\pi}_{bqs}, \tag{26}$$

where $\boldsymbol{Y}_{bd}$ is the regression matrix of fully dynamic parameters $\boldsymbol{\pi}_{bd}$, $\boldsymbol{Y}_{bf}$ is the regression matrix of friction parameters $\boldsymbol{\pi}_{bf}$. The relationship between the expanded regression matrix $\boldsymbol{Y}_{b2}$ and fully dynamic parameters $\boldsymbol{\pi}_{b2}$ can be expressed as

$$\boldsymbol{Y}_{b2}\boldsymbol{\pi}_{b2} = \boldsymbol{J}_h\boldsymbol{P}_L - \boldsymbol{Y}_{bqs}\hat{\boldsymbol{\pi}}_{bqs}, \tag{27}$$

where $\boldsymbol{Y}_{b2} = [\boldsymbol{Y}_{bd} \quad \boldsymbol{Y}_{bf}]$, $\boldsymbol{\pi}_{b2} = [\boldsymbol{\pi}_{bd}^T \quad \boldsymbol{\pi}_{bf}^T]^T$. $\boldsymbol{Y}_{bf}$ is the regression matrix of friction parameters $\boldsymbol{\pi}_{bf}$, which can be written as

$$\boldsymbol{Y}_{bf} = \text{diag}\left[\boldsymbol{Y}_{bf,1},\ldots,\boldsymbol{Y}_{bf,i},\ldots,\boldsymbol{Y}_{bf,n}\right], \tag{28}$$

where $\boldsymbol{Y}_{bf,i} = \text{diag}[\dot{q}_i, sign(\dot{q}_i), 1]$. The observation matrix $\boldsymbol{\Phi}_{b2}$ for s rounds of data collection can be written as

$$\boldsymbol{\Phi}_{b2} = \begin{bmatrix}\boldsymbol{Y}_{b2,1}^T & \boldsymbol{Y}_{b2,2}^T & \cdots & \boldsymbol{Y}_{b2,s}^T\end{bmatrix}^T, \tag{29}$$

The sampled torque signal $\boldsymbol{\omega}_{b2}$ can be written as

$$\boldsymbol{\omega}_{b2} = \begin{bmatrix}\boldsymbol{J}_h\boldsymbol{P}_{L,1} - \boldsymbol{Y}_{bqs2,1}\hat{\boldsymbol{\pi}}_{bqs}\\ \boldsymbol{J}_h\boldsymbol{P}_{L,2} - \boldsymbol{Y}_{bqs2,2}\hat{\boldsymbol{\pi}}_{bqs}\\ \vdots\\ \boldsymbol{J}_h\boldsymbol{P}_{L,s} - \boldsymbol{Y}_{bqs2,s}\hat{\boldsymbol{\pi}}_{bqs}\end{bmatrix}. \tag{30}$$

The regression equation can be written as

$$\boldsymbol{\omega}_{b2} = \boldsymbol{\Phi}_{b2}\boldsymbol{\pi}_{b2}. \tag{31}$$

To fully excite fully dynamic parameters and friction parameters, the excitation trajectory is designed as a combination of Fourier series and polynomial functions as

$$q_{2,i}(t) = \sum_{l=1}^{7}\left(\frac{a_{2,i,l}}{\omega_{2f}}\sin(\omega_{2f}lt) - \frac{b_{2,i,l}}{\omega_{2f}}\cos(\omega_{2f}lt)\right) + c_{2,i,0} + \sum_{l=1}^{5}c_{2,i,l}t^l \quad i = 1,\ldots,n, \tag{32}$$

$$\text{minimize } \|\text{cond}(\boldsymbol{\Phi}_{b2}(\boldsymbol{q},\dot{\boldsymbol{q}},\ddot{\boldsymbol{q}}))\|, \tag{33}$$

where $a_{2,i,l}$, $b_{2,i,l}$, $c_{2,i,l}$, and $c_{2,i,0}$ represent coefficients of Fourier series and polynomial functions serving as optimization variables, ω_{2f} corresponding to the base frequency of Fourier series is related to the period of the trajectory.

A physical feasibility set for fully dynamic parameters is established as

$$\mathcal{B}_d \equiv \{\boldsymbol{\pi}_{bd} \in \mathbb{R}^{n_b - n_{qs}} : \exists \boldsymbol{\pi}_d \in \mathbb{R}^{n_d} | \boldsymbol{D}_b\left(\left[\hat{\boldsymbol{\pi}}_{bqs}^T \quad \hat{\boldsymbol{\pi}}_{bd}^T\right]^T, \boldsymbol{\pi}_d\right) \succ 0\}, \tag{34}$$

where n_{qs} is the number of quasi-static parameters. And a semidefinite programming approach is utilized to estimate fully dynamic parameters and friction parameters that satisfy physical feasibility as

$$\text{minimize } \left\|\boldsymbol{\Phi}_{b2}\left[\hat{\boldsymbol{\pi}}_{bd}^T \quad \hat{\boldsymbol{\pi}}_{bf}^T\right]^T - \boldsymbol{\omega}_{b2}\right\|, \quad \text{s.t. } \hat{\boldsymbol{\pi}}_{bd} \in \mathcal{B}_d, \hat{f}v_i > 0, \hat{f}c_i > 0. \tag{35}$$

IV. Experiment Procedure

A. Test Bench

As depicted in Figs. 1 and 3, the test bench includes a 7-DOF hydraulic manipulator, which is actuated by a combination of linear and swing cylinders, and regulated by servo valves. Pressure sensors and angular encoders are equipped to measure cylinder pressures and joint angles, respectively. A real-time control system is established to gather pressure and angle signals via the data acquisition device with a sampling frequency of 1 kHz. All

Fig. 3. Hydraulic manipulator used for experiment.

TABLE I
THE CONSTRAINTS FOR EXCITATION TRAJECTORIES

Parameters	Value
joint limits (i=1)	-0.6981 to 0.5236 /rad
joint limits (i=2)	0.1745 to 1.9199 /rad
joint limits (i=3)	-0.5236 to 0.5236 /rad
velocity limits	-0.25 to 0.25 /rad·s^{-1}
acceleration limits	-1 to 1 /rad·s^{-2}

algorithms were executed utilizing MATLAB/Simulink Real-Time. The shoulder pitch, elbow pitch, and wrist pitch joints of the manipulator are used, which can be considered as a 3-DOF manipulator driven by three linear cylinders. The main structure parameters can be found in [23], [31]. All dynamic parameters can be expressed as

$$\boldsymbol{\pi}_{bqs} = \begin{bmatrix} h_{y,3} \\ h_{x,3} \\ h_{y,2} \\ h_{x,2} + m_3 d_4 \\ h_{y,1} + a_2(m_2 + m_3) \\ h_{x,1} + d_2(m_2 + m_3) \end{bmatrix}, \tag{36}$$

$$\boldsymbol{\pi}_{bd} = \begin{bmatrix} I_{zz,3} \\ I_{zz,2} + d_4^2 m_3 \\ I_{zz,1} + (a_2^2 + d_2^2)(m_2 + m_3) \end{bmatrix}, \tag{37}$$

$$\boldsymbol{\pi}_{bf} = [fv_1, fc_1, fo_1, fv_2, fc_2, fo_2, fv_3, fc_3, fo_3]^T, \tag{38}$$

where $I_{zz,i}$ represents the inertia of the ith link around the Z-axis of the ith joint coordinate system.

B. Excitation Trajectories for Sequential Identification

According to excitation trajectories expressed in (24) and (32), their coefficients are mainly determined based on the condition number of the observation matrix. The corresponding optimization problems have been expressed in (23) and (33), respectively. To achieve more accurate identification, multiple sets of optimal coefficients have been generated under the same motion constraints, shown in Table I. Then, the set with the minimum condition number is selected for the identification. The selected excitation trajectory for quasi-static parameters is depicted in Fig. 4(a), and its coefficients are shown in Table II. The selected excitation trajectory for fully dynamic parameters and friction parameters is depicted in Fig. 4(b), and its coefficients are shown in Table III.

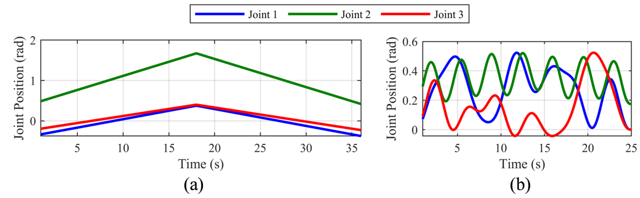

Fig. 4. Excitation trajectories designed for the proposed method. (a) Trajectory $q_i(t)$ for identifying quasi-static parameters $\boldsymbol{\pi}_{bqs}$, condition number: 12; (b) Trajectory $q_{2,i}(t)$ for identifying fully dynamic parameters $\boldsymbol{\pi}_{bd}$ and friction parameters $\boldsymbol{\pi}_{bf}$, condition number: 7.

TABLE II
COEFFICIENTS OF THE EXCITATION TRAJECTORY FOR QUASI-STATIC PARAMETERS

Joint i	$q_{1,i,0}$ /rad	$q_{1,i,f}$ /rad	T /s
1	-0.37	0.38	
2	0.42	1.67	18
3	-0.23	0.40	

TABLE III
COEFFICIENTS OF THE EXCITATION TRAJECTORY FOR FULLY DYNAMIC PARAMETERS AND FRICTION PARAMETERS

Joint i	$a_{2,i,1}$	$a_{2,i,2}$	$a_{2,i,3}$	$a_{2,i,4}$	$a_{2,i,5}$	$a_{2,i,6}$	$a_{2,i,7}$
1	17.97	1.18	0.21	-0.06	0.06	0.01	0.08
2	19.29	1.2	0.23	0.09	0.02	0.02	0.02
3	8.42	0.48	0.13	0.13	0.01	0.03	-0.07
-	$b_{2,i,1}$	$b_{2,i,2}$	$b_{2,i,3}$	$b_{2,i,4}$	$b_{2,i,5}$	$b_{2,i,6}$	$b_{2,i,7}$
1	8.49	1.06	0.38	0.15	0.13	0.08	0.06
2	14.36	1.8	0.54	0.23	0.13	0.09	0.28
3	8.25	1.06	0.36	0.19	0.09	0.09	0.06
-	$c_{2,i,0}$	$c_{2,i,1}$	$c_{2,i,2}$	$c_{2,i,3}$	$c_{2,i,4}$	$c_{2,i,5}$	ω_{2f}
1	36.75	-19.45	-1.75	0.45	-0.02	0	
2	62.14	-20.88	-2.97	0.57	-0.02	0	0.25
3	35.78	-9.13	-1.71	0.28	-0.01	0	

C. Excitation Trajectories for Comparative Methods

The design of the excitation trajectory for the comparative methods is conducted with the same criterion as illustrated in Section IV-B. All dynamic parameters of the hydraulic manipulator are simultaneously identified. While the optimization objectives depend on the detailed method, all constraint conditions, such as joint limits, velocity limits, and acceleration limits, are kept consistent across all methods. By maintaining these consistent constraints, the parameter identification results are directly comparable across different methods, reinforcing the integrity of the experiment design and the validity of the comparison. To ensure rationality, the same excitation trajectory is used for the identification of all comparative methods. Eight excitation trajectories were generated, and the one with the smallest condition number is selected for the comparative experiment. Its condition number is 343 as shown in Fig. 5, and its coefficients are depicted in Table IV.

Note that some parameters related to the inertia term can not be sufficiently excited when all dynamic parameters are simultaneously excited. In other word, while optimizing the excitation trajectory with the same low-speed motion constraints, condition numbers in comparative methods tend to increase. Thus, there is

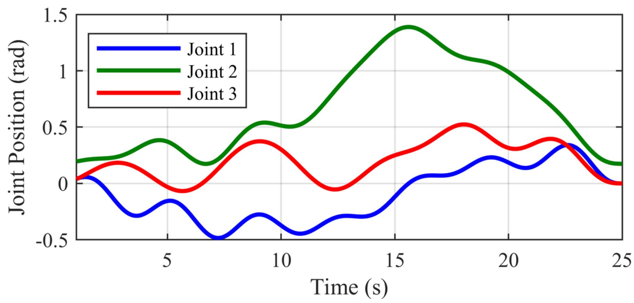

Fig. 5. The excitation trajectories for comparative methods.

TABLE IV
COEFFICIENTS OF THE EXCITATION TRAJECTORY FOR COMPARATIVE METHODS

Joint i	$a_{o,i,1}$	$a_{o,i,2}$	$a_{o,i,3}$	$a_{o,i,4}$	$a_{o,i,5}$	$a_{o,i,6}$	$a_{o,i,7}$
1	18.69	1.17	0.23	0.05	0.03	0.08	0.05
2	16.76	1.12	0.19	0.09	0	0.05	0.04
3	14.33	0.87	0.23	0.06	-0.02	0.05	0.03
-	$b_{o,i,1}$	$b_{o,i,2}$	$b_{o,i,3}$	$b_{o,i,4}$	$b_{o,i,5}$	$b_{o,i,6}$	$b_{o,i,7}$
1	8.27	1.05	0.3	0.15	0.07	0.07	0.15
2	3.78	0.51	0.16	0.13	-0.01	-0.01	0.06
3	4.25	0.58	0.11	0.17	0.07	0.05	-0.02
-	$c_{o,i,0}$	$c_{o,i,1}$	$c_{o,i,2}$	$c_{o,i,3}$	$c_{o,i,4}$	$c_{o,i,5}$	ω_o
1	35.75	-20.3	-1.72	0.46	-0.02	0	
2	16.58	-18.25	-0.76	0.35	-0.02	0	0.25
3	18.47	-15.55	-0.88	0.32	-0.02	0	

such a significant difference in condition numbers between the proposed method and other comparative.

The combination $\boldsymbol{\pi}_o$ of base inertial parameters and friction parameters can be expressed as

$$\boldsymbol{\pi}_o = \left[\boldsymbol{\pi}_{bqs}^T \quad \boldsymbol{\pi}_{bd}^T \quad \boldsymbol{\pi}_{bf}^T\right]^T. \tag{39}$$

The combination of the regression matrix for base inertial parameters and the regression matrix for friction parameters can be expressed as

$$\boldsymbol{Y}_o = \left[\boldsymbol{Y}_{bqs} \quad \boldsymbol{Y}_{bd} \quad \boldsymbol{Y}_{bf}\right]. \tag{40}$$

The observation equation can be written as

$$\boldsymbol{\Phi}_o = \left[\boldsymbol{Y}_{o,1}^T \quad \boldsymbol{Y}_{o,2}^T \quad \cdots \quad \boldsymbol{Y}_{o,s}^T\right]^T, \tag{41}$$

$$\boldsymbol{\omega}_o = \left[\boldsymbol{J}_{h,1}\boldsymbol{P}_{L,1} \quad \boldsymbol{J}_{h,2}\boldsymbol{P}_{L,2} \quad \cdots \quad \boldsymbol{J}_{h,2}\boldsymbol{P}_{L,s}\right]^T, \tag{42}$$

$$\boldsymbol{\omega}_o = \boldsymbol{\Phi}_o\boldsymbol{\pi}_o. \tag{43}$$

The excitation trajectory for comparative methods is generated by minimizing the condition number of matrix $\boldsymbol{\Phi}_o$ as

$$q_{o,i}(t) = \sum_{l=1}^{7}\left(\frac{a_{o,i,l}}{\omega_o}\sin(\omega_o lt) - \frac{b_{o,i,l}}{\omega_o}\cos(\omega_o lt)\right) + c_{o,i,0} + \sum_{l=1}^{5} c_{o,i,l}t^l \; i = 1,\ldots,n, \tag{44}$$

$$\text{minimize} \left\|\text{cond}(\boldsymbol{\Phi}_o(\boldsymbol{q},\dot{\boldsymbol{q}},\ddot{\boldsymbol{q}}))\right\|. \tag{45}$$

D. Trajectories for Validation

In the process of validation, 12 trajectories are generated and their motion constraints remain consistent with all excitation trajectories shown in Table I. These trajectories are grouped into 2 sets, where 8 trajectories are designed using the combination of Fourier series and polynomials shown in (44), and 4 trajectories are designed using the constant velocity trajectory shown in (22). Their labels are depicted in Table V.

TABLE V
LABELS OF 12 VALIDATION TRAJECTORIES

Trajectories	Labels
Fourier series and polynomials	A, B, C, D, E, F, G, H
Constant velocity	H, I, J, K

E. Data Processing

A consistent data processing approach is used for different identification methods to ensure the validity of comparisons. Joint trajectories are tracked by a PID controller. Concurrently, pressure signals and joint angle signals are recorded. The angular velocity, angular acceleration, and pressure differentiation are determined through numerical differentiation. The measured data must be processed to mitigate the impact of noise. Given that the data are processed offline, a zero-phase filter is employed to minimize the high-frequency noise generated by numerical differentiation. Specifically, a 3rd-order low-pass Butterworth filter with a cutoff frequency of 2.8 Hz is utilized, derived from 10 times the base frequency over $10\times 7\omega_f/2\pi = 2.8$ Hz, which may effectively attenuate the high-frequency measurement noise.

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. Parameter Identification Method Used for Comparison

Identification result in (25) and (35) of the proposed method can be written as

$$\hat{\boldsymbol{\pi}}_{S,PF} = \left[\hat{\boldsymbol{\pi}}_{bqs}^T \quad \hat{\boldsymbol{\pi}}_{bd}^T \quad \hat{\boldsymbol{\pi}}_{bf}^T\right]^T. \tag{46}$$

The least squares method (LS) is utilized in parameter identification. This method seeks to minimize the sum of the squared differences between observed and predicted torque values. By solving a system of linear equations, the least squares estimation of all parameters are obtained as

$$\hat{\boldsymbol{\pi}}_{o,LS} = \left(\boldsymbol{\Phi}_o^T\boldsymbol{\Phi}_o\right)^{-1}\boldsymbol{\Phi}_o^T\boldsymbol{\omega}_o. \tag{47}$$

The weighted least squares method (WLS) is a refinement of the base least squares approach, specifically tailored to handle scenarios where the data points may not hold the same degree of reliability. It works by assigning different weights to data points in the cost function, allowing for a more nuanced adjustment of the model to the observed data. The weights typically reflect the variance of each observation, leading to more accurate parameter estimation in the presence of heteroscedasticity or unequal noise levels as

$$\sigma^2(i) = \frac{\left\|\boldsymbol{\omega}_o(k) - \boldsymbol{\Phi}_o(k)\hat{\boldsymbol{\pi}}_{o,LS}\right\|^2}{s - n_b}, \tag{48}$$

where $\hat{\sigma}$ is the variance of the observation error, $k = i, i+n,\ldots,s-n+1$. The weight matrix $\boldsymbol{W}$ is given by

$$\boldsymbol{W} = \begin{bmatrix} diag(\boldsymbol{W})_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & diag(\boldsymbol{W})_s \end{bmatrix}, \tag{49}$$

TABLE VI
THE IDENTIFICATION RESULTS

Parameters	Proposed	LS	WLS	PF
$\hat{\boldsymbol{\pi}}_{bqs}(1)$	1.5664	2.8912	2.5855	3.1162
$\hat{\boldsymbol{\pi}}_{bqs}(2)$	32.6657	30.7708	31.5989	30.5863
$\hat{\boldsymbol{\pi}}_{bqs}(3)$	1.9426	0.4345	0.0015	0.1123
$\hat{\boldsymbol{\pi}}_{bqs}(4)$	36.4658	41.7688	41.1202	41.7798
$\hat{\boldsymbol{\pi}}_{bqs}(5)$	112.0147	122.2384	122.3653	119.1893
$\hat{\boldsymbol{\pi}}_{bqs}(6)$	146.3663	151.6838	150.7552	148.8312
$\hat{\boldsymbol{\pi}}_{bd}(1)$	14.2865	14.421	13.656	8.1946
$\hat{\boldsymbol{\pi}}_{bd}(2)$	23.4857	-0.5924	-2.679	15.0098
$\hat{\boldsymbol{\pi}}_{bd}(3)$	111.8606	143.8879	150.7552	141.7154
$\hat{f}v_1$	57.4165	57.38	146.6044	59.9037
$\hat{f}c_1$	119.9133	105.6218	43.917	105.4959
$\hat{f}o_1$	7.9706	-117.0201	106.15	-75.369
$\hat{f}v_2$	15.8395	145.6268	-112.3097	141.5711
$\hat{f}c_2$	48.4948	27.2244	146.8386	27.4929
$\hat{f}o_2$	-5.949	-43.3948	28.1336	-41.5493
$\hat{f}v_3$	13.7858	8.1131	-43.9614	8.9439
$\hat{f}c_3$	12.5067	12.8093	14.1451	12.911
$\hat{f}o_3$	4.4977	18.8202	12.1892	20.3284

where $\boldsymbol{W} = 1/[\sigma(1), \sigma(2), \ldots, \sigma(n)]$.

The estimation result of weighted least squares is given by

$$\hat{\boldsymbol{\pi}}_{o,WLS} = \left(\boldsymbol{W}\boldsymbol{\Phi}_o^T\boldsymbol{W}\boldsymbol{\Phi}_o\right)^{-1}\boldsymbol{W}\boldsymbol{\Phi}_o^T\boldsymbol{W}\boldsymbol{\omega}_o. \tag{50}$$

The parameter identification method only considering physical feasibility (PF) is aiming at ensuring that the identified parameters satisfy constraints of physical feasibility. This method is usually used in the presence of complex physical environments and sensor limitations. By introducing physical constraints, this method seeks to enhance the reliability and practicality of the identification results as

$$\begin{aligned} &\text{minimize } \left\|\boldsymbol{\Phi}_o\hat{\boldsymbol{\pi}}_{o,PF} - \boldsymbol{\omega}_o\right\|, \\ &\text{s.t. } \hat{\boldsymbol{\pi}}_{o,PF} \in \mathcal{B}, \hat{f}v_i > 0, \hat{f}c_i > 0. \end{aligned} \tag{51}$$

B. Identification Results

The identification results are shown in Table VI. The first column of the table presents the results obtained by the proposed method, followed by the results of LS, WLS, and PF in the second, third, and fourth columns, respectively. The parameters listed from top to bottom are quasi-static parameters, fully dynamic parameters, and friction force parameters, as shown in (39).

In the identification results, the physical infeasibility occurred with LS and WLS, as shown in the data highlighted with a red background in Table VI, while it did not occur with the proposed method and PF. For the results of LS, WLS, and PF, the offset term of the friction model is significant, even exceeding the Coulomb term, as shown in the data highlighted with an orange background in Table VI. The reason is that when quasi-static parameters and friction parameters are simultaneously identified, the damping friction becomes less dominant. Then, the disturbance term of the friction are more sensitive and the observation matrix becomes ill-conditioned. In such cases, estimation results of these highlighted parameters may deviate from true values and present significant difference compared with the value of the proposed method. From the results using the proposed method, this phenomenon is avoided owing to the excitation trajectory with smaller condition number, which is designed in Section IV-B. This sequential identification method reduces the mutual interference caused by noise in different parameters.

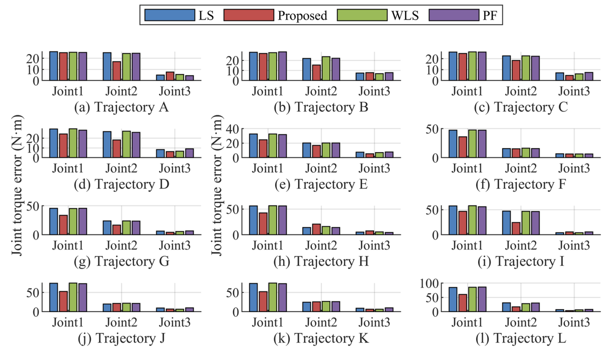

Fig. 6. Mean torque prediction errors for all validation trajectories using the proposed method and comparative methods.

TABLE VII
RMS OF THE TORQUE PREDICTION ERROR AND RATIOS BETWEEN THE RMS OF THE PROPOSED METHOD AND THAT OF COMPARATIVE METHODS

Trajectory	Proposed	LS		WLS		PF	
	RMS	RMS	r_{rms}	RMS	r_{rms}	RMS	r_{rms}
A	25.29	27.7	91.30%	27.65	91.47%	27.24	92.84%
B	26.2	27.86	94.05%	28.34	92.44%	27.98	93.62%
C	25.2	26.83	93.94%	26.94	93.56%	26.71	94.37%
D	25.09	29.96	83.72%	30.02	83.57%	29.34	85.51%
E	24.55	29.23	83.99%	29.11	84.33%	28.75	85.39%
F	30.84	34.86	88.45%	34.92	88.30%	34.83	88.55%
G	29.86	36.53	81.74%	36.24	82.4%	36.45	81.93%
H	36.6	39.88	91.77%	40.18	91.09%	39.79	92.00%
I	33.46	44.17	75.76%	43.86	76.30%	43.06	77.71%
J	38.67	50.04	77.26%	50.57	76.46%	50.22	77.00%
K	39.6	50.67	78.15%	51.28	77.22%	50.88	77.83%
L	40.46	55.02	73.54%	55.28	73.19%	55.47	72.94%

C. Torque Prediction Accuracy

The results of dynamic parameters identification are then utilized to predict joint torques under 12 validation trajectories, designed in Section IV-D. The mean torque prediction error is selected to describe the torque prediction accuracy of all methods in Fig. 6. The ground truth values for obtaining errors are actual joint torques, which are calculated based on the output force of hydraulic cylinders through measuring pressure signals [23], [32]. In addition, the root mean square (RMS) value of the torque prediction error has also been calculated in Table VII. A ratio r_{rms} between the RMS of the proposed method and that of each comparative method is defined, to assess the extent of performance improvement achieved by the proposed method compared to comparative methods. As r_{rms} is less than 1, it indicates that the proposed method yields a smaller prediction error and achieves higher prediction accuracy. When the ratio decreases, the extent of performance improvement increases.

As shown in Table VII, the proposed method shows significantly lower prediction errors. The quasi-static parameters account for a significant portion of the predicted torque, highlighting that accurate identification of quasi-static parameters is crucial for heavy-duty hydraulic manipulators. The RMS value of the torque prediction error is the smallest using the proposed method, demonstrating the improvement relative to other methods due to the accurate identification of dynamic parameters. Although there is no quantitative analysis on the sensitivity of the proposed sequential identification method to measurement noise, the results indirectly illustrate that the proposed method inherently enhances the robustness against measurement noise. Additionally, the constant velocity validation trajectory exhibits better performance compared with other methods, which can be attributed to the more stable friction during the constant velocity phase. In summary, according to the Fig. 6 and Table VII, more accurate dynamic parameters can be identified by the proposed method, so a more accurate torque prediction result is achieved under various working conditions.

VI. Conclusion

In this letter, the parameter identification accuracy of heavy-duty hydraulic manipulators is further improved. A sequential parameter identification approach is proposed, which divides dynamic parameters into two groups, each excited using different excitation trajectories. Each group of inertia parameters is fully excited (the condition number of the observation matrix is significantly reduced compared to traditional methods), due to the division of dynamic parameters into groups. Based on the collected data, two groups of parameters are estimated within the physically feasible set. The proposed method fully excites all dynamic parameters and ensures their physical feasibility. A detailed evaluation of the proposed method has been conducted, with performance comparisons made against various identification methods, including least squares, weighted least squares, and the method only considering physical feasibility. The results reveal a significant reduction in torque prediction errors, and the prediction accuracy of the joint torque using the proposed method has been improved by approximately 5.63% to 27.06%, compared with comparative methods.

With this novel and effective approach, an accurate dynamic model of the heavy-duty hydraulic manipulator can be established, which is essential to achieve the accurate model-based motion control. It makes accurate perception of external contact forces possible in the human-robot-environment interaction. Future work will focus on online payload estimation and contact force control of heavy-duty hydraulic manipulators.

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